Sri Lanka Institute of Information Technology



IT3091 – MACHINE LEARNING

Model / Method Comparison

Group Number: 2026-DS-35

Student ID	Student Name
IT24104087	Rathnamalala R.I.B.T
IT24103907	Pamindi M.M.
IT24400066	Athauda P.A.K.D.
IT24103888	Gunathunga K.S.N.T.

Model	Method	Accuracy	Precision	Recall	F1	ROC AUC	CV Accuracy	Interpretation
Logistic Regression	Baseline linear classifier	0.7227	0.7902	0.6729	0.7269	0.8028	0.7378 ± 0.0022	Simple and interpretable, but weaker on non-linear interactions.
Decision Tree	Single rule-based tree	0.7644	0.8209	0.7295	0.7725	0.8485	0.7043 ± 0.0048	Strong test result and easy to explain; single trees can be less stable.
Random Forest	Bagged tree ensemble	0.7601	0.8436	0.6904	0.7594	0.8540	0.7355 ± 0.0032	Highest precision; robust ensemble and useful backup option.
XGBoost	Boosted tree ensemble	0.7742	0.8347	0.7335	0.7808	0.8681	0.7401 ± 0.0026	Best test Accuracy, F1 and ROC AUC; notebook's recommended operational model.

Method comparison and final selection

Logistic Regression provides a transparent baseline but has the lowest test accuracy and ROC AUC, indicating that a purely linear decision boundary does not capture all delay-risk interactions.

Decision Tree improves substantially over the baseline and is useful for explaining human-readable rules. Its cross-validation mean is lower than its held-out test accuracy, so it is best treated as an interpretable supporting model rather than the main deployment choice.

Random Forest produces the highest precision (0.8436), meaning that among orders it flags as late, a high proportion are actually late. Its recall is lower than the Decision Tree and XGBoost.

XGBoost has the strongest overall held-out performance: Accuracy 0.7742, F1 0.7808 and ROC AUC 0.8681. The notebook therefore recommends XGBoost as the primary model, with Random Forest as a backup and the Decision Tree as an explainability aid.

The cross-validation results are close with low standard deviations, supporting reasonably stable model behaviour across folds. The CV ordering differs somewhat from the held-out test ordering, so the recommendation is based on the full set of test metrics rather than accuracy alone.